Soft Sensor for Product Quality Monitoring in a Debutanizer Distillation Column

Project Report

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Process Description

The Debutanizer Column

A debutanizer is a distillation column widely used in petroleum refining and natural-gas processing. Its primary function is to separate butane (C4 hydrocarbons) from heavier components such as isopentane (iC5) and pentane (C5) in a liquid feedstock. Feed enters the column mid-section; lighter C4 components rise as vapour and exit from the top as an overhead product, while heavier C5+ components sink and leave from the bottom as the main product stream.

The industrial column studied in Fortuna et al. (2005) belongs to the ERGMED refining plant. It operates continuously, and the quality of the bottom product is characterised by the fraction of C4 remaining in the iC5 stream. Keeping this fraction low is the primary control objective, because excess C4 degrades the quality of the downstream liquid product.

Process Flowsheet and Variable Mapping

Figure 1 summarises the column's physical layout and the measurement points associated with each process variable.

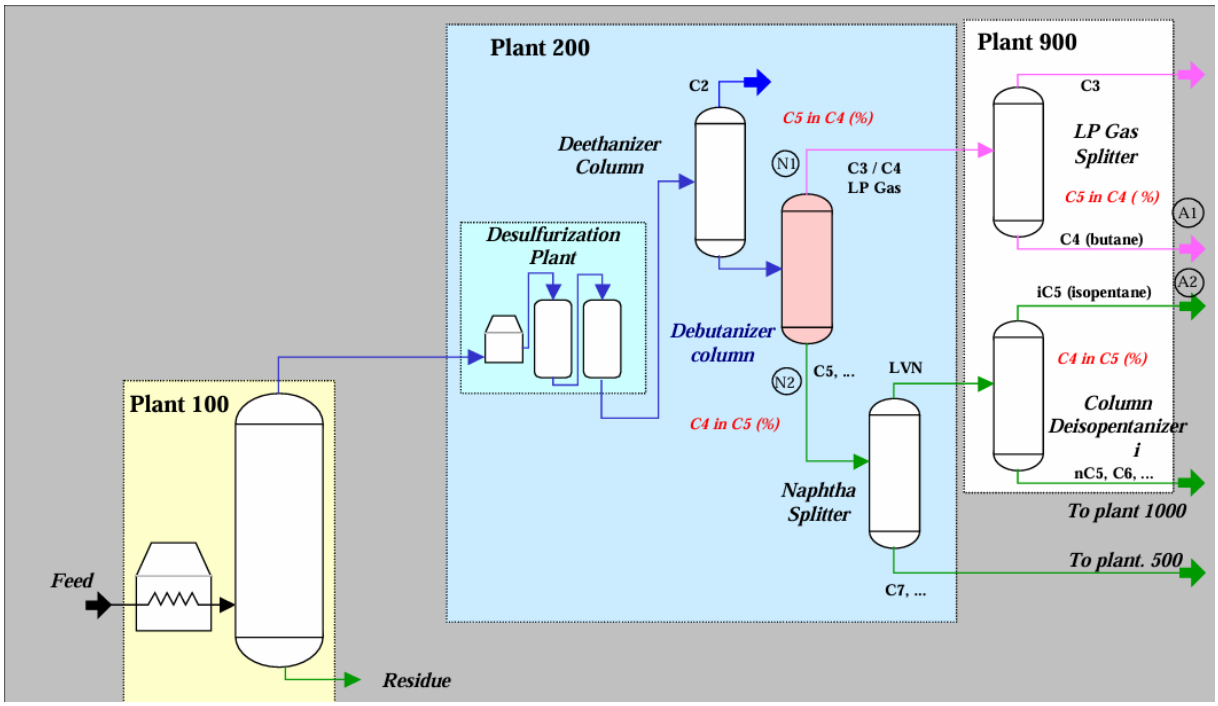


Figure 1. Process flowsheet of the debutanizer distillation column, showing the column layout, major equipment (reboiler E108AB, condenser E107AB, pump P102AB), and the measurement points corresponding to each process variable (T040, P011, F015, F018, T004, T036, T037, A002BR).

The feed enters the column mid-section and heat is supplied at the bottom via the reboiler (E108AB), which vaporises lighter components upward. The overhead vapour is condensed in condenser E107AB, partially returned as reflux via pump P102AB, and partially drawn off as the overhead product (C4-rich stream) to the LPG splitter in Plant 900. The bottom product (iC5-rich stream) exits and flows to the deisopentanizer for further processing.

The seven measured variables map directly onto this physical structure:

- **u1 (T040) – Top temperature.** Measured at the column overhead. Indicates the composition of the vapour leaving the column. A rise in top temperature signals that heavier components (C5) are breaking through into the overhead, while a lower temperature indicates the overhead is dominated by the lighter C4 fraction.
- **u2 (P011) – Top pressure.** The column operates under a controlled pressure setpoint. Pressure directly affects the vapour-liquid equilibrium: higher pressure raises the bubble point of the mixture, altering the relative volatilities and therefore the separation efficiency. Pressure fluctuations disturb the steady-state composition profile throughout the column.
- **u3 (F015) – Reflux flow rate.** The single most important operator handle for controlling top-product purity. Reflux is the condensed overhead liquid returned to the column top. A higher reflux ratio washes heavier C5 components back down the column, improving C4 purity in the overhead and reducing residual C4 in the bottoms. Reducing reflux saves energy but allows C5 to escape upward and permits more C4 to remain in the bottom product.

- **u4 (F018) – Feed flow to Plant 900 (overhead product draw).** The rate at which the overhead product is withdrawn. Together with reflux, this determines the reflux ratio L/V. Changes in the draw rate directly alter the column loading and shift the separation tray by tray.
- **u5 (T004) – VIth tray temperature.** Measured at an intermediate tray in the middle section of the column. This is a classic inferential measurement in distillation practice: the axial temperature profile tracks the composition profile, so a shift in the tray temperature indicates that the C4/C5 cut point has moved up or down the column. The VIth tray is particularly sensitive to the butane–isopentane separation and responds within 1–2 sampling periods to a reflux change, making it an early indicator of future changes in bottom product quality.
- **u6 (T036), u7 (T037) – Bottom temperatures (E108A and E108B).** Measured at the two sides of the reboiler. These indicate the thermal duty being delivered to the column bottom. A higher bottom temperature implies greater boilup, which strips lighter components upward and drives the C4 content of the bottom product lower. The two sensors respond in tandem to the same reboiler duty change; the paper averages them as $u_{\text{avg}} = (T036 + T037)/2$ to obtain a more robust single measure of the effective reboiler duty.
- **y (A002BR) – C4 content in the iC5 bottom product.** The target quality variable, measured by the gas chromatograph on the deisopentanizer overhead with a sampling cycle of approximately 15 min and a total transport plus analysis delay of 30–75 min. This is the variable the soft sensor estimates in real time, bypassing the analyser dead time.

Why These Variables Predict y

The bottom product composition is governed by the column's coupled energy and mass balances. Any disturbance in reflux, feed rate, reboiler duty, or pressure propagates through the column as a temperature wave – observable at T040 (top), T004 (middle tray), and T036/T037 (bottom) – before eventually manifesting as a composition change in the bottom stream. The NARMAX model captures these lagged cause-and-effect relationships by including time-shifted versions of the process measurements as model inputs.

Separation Mechanism and Reflux Dynamics

Thermodynamic Basis for Separation

The driving force for separation in the debutanizer is the volatility difference between C4 (butane, normal boiling point −1 °C) and iC5 (isopentane, boiling point 28 °C). The column operates as a multi-stage countercurrent contactor: hot vapour rises from the reboiler (bottom temperature approximately 70–80 °C) while cool liquid descends from the condenser. At each tray, rising vapour and descending liquid reach vapour-liquid equilibrium, progressively enriching the vapour in C4 (more volatile) and the liquid in C5+ (less volatile) with each successive stage.

Effect of Reflux on Bottom Composition

The reflux ratio $R = L/D$ (liquid returned to the column / distillate withdrawn) is the primary control handle. Consider the effect of a step increase in reflux:

1. Additional cold liquid enters the top of the column, cooling the upper trays.
2. The increased liquid load cascades downward, providing more liquid surface for vapour–liquid exchange on each tray.
3. Heavier iC5 components are more effectively absorbed into the descending liquid and washed back toward the bottoms.
4. Simultaneously, C4 is stripped more efficiently into the rising vapour and carried out of the bottom product.
5. The net result is a decrease in C4 concentration in the bottom product y, with the VIth tray temperature (u5) dropping as the composition front moves downward.

The inverse relationship – ↑ reflux (u3) ⇒ ↓ C4 in bottoms (y) – is confirmed by the negative cross-correlation between u3 and y in the temporal analysis (Section 4).

Origin of the Lag Structure

A step change in reflux at time k does not produce an instantaneous change in bottom product composition. Instead, a cascade of delayed responses unfolds:

- The VIth tray temperature (u5 = T004) responds within 1–2 sampling periods (15–30 min), because it sits in the middle of the column where the temperature gradient is steepest and the composition front arrives early.
- The bottom temperatures (u6, u7 = T036, T037) respond slightly later, as the thermal disturbance must travel further down before reaching the reboiler section.
- The bottom composition y responds last: after the new liquid distribution has re-established vapour-liquid equilibrium throughout all column stages. This final equilibration typically takes 45–90 min, corresponding to 3–6 lags at $T_s = 15$ min.

This physical picture explains directly why the CCF between u3 and y shows peak correlation at lags 3–6, why the NARMAX model includes T004 lagged up to 3 steps (T004 is an early indicator of what y will be 2–3 samples later), and why a static model using only instantaneous measurements is structurally incapable of capturing the dominant predictive relationships in this process.

Role of the Reboiler Temperatures

The reboiler temperatures u6 and u7 control the boilup rate V. Higher reboiler duty increases V, raising the vapour-to-liquid ratio on the lower trays, which strips more C4 from the liquid phase and carries it up the column – thereby reducing C4 in the bottom product. The paper averages T036 and T037 because both measure the same reboiler duty from complementary points; their average is a more robust measure of the actual thermal input than either sensor alone, and is confirmed by a Pearson correlation of $r = 0.99$ between the two.

Role of Column Pressure

Higher column pressure (u2 = P011) shifts the vapour-liquid equilibrium by raising the bubble points of all components. At fixed temperatures, this tends to suppress vaporisation and push heavier components into the vapour phase. In practice, however, the

column operates at a tightly controlled pressure setpoint and u2 varies very little (standard deviation 0.058 compared to 0.13–0.20 for the temperature variables). Its CCF with y is correspondingly flat, making pressure a minor contributor to the soft sensor signal.

Why Quality Measurement Is Difficult

The standard instrument for measuring bottom-product quality is a gas chromatograph (GC), which draws a liquid sample, separates it by component, and reports concentrations. The GC measurement is authoritative, but it has two critical limitations:

- **Measurement delay.** The complete cycle of sampling, separation, and reporting takes between 20 and 75 minutes. For the specific configuration studied, the effective delay is approximately $\Delta t = 45$ min, which corresponds to 3 sampling intervals at the chosen sampling period of $T_s = 15$ min.
- **High cost and low availability.** A chromatograph dedicated to a single column is expensive, requires specialist maintenance, and is prone to downtime. Many plants share one GC across multiple streams.

Because the control loop cannot wait 45 minutes for a quality measurement, operators are forced to use slower setpoint adjustments or rely on their experience. This motivates the idea of a *soft sensor*: a mathematical model that infers product quality in real time from inexpensive, high-frequency measurements already available from plant instrumentation.

The Soft Sensor Concept

A soft sensor replaces or supplements the physical analyser by taking signals that are measured cheaply and continuously (temperatures, pressures, flow rates) and computing an estimate of the quantity of interest (the C4 concentration). Because these secondary measurements are available every sampling period without delay, the soft sensor can provide a real-time quality estimate to the control system, eliminating the 45-minute dead time.

The paper by Fortuna et al. (2005) proposes a neural-network-based soft sensor trained on three years of historical plant data. The architecture chosen is a NARMAX model (Nonlinear AutoRegressive Moving Average with eXogenous inputs) implemented as a multilayer perceptron (MLP), because the concentration dynamics are nonlinear and have significant temporal memory.

Dataset Description

Source and Pre-processing

The dataset used in this project is a publicly available pre-processed version of the industrial data described in Fortuna et al. (2005). The original three-year raw dataset from the ERGMED plant was smoothed, normalised, and reduced to 2394 samples before release. All values have been min-max scaled to $[0, 1]$, so no additional normalisation is required.

Variables

The dataset contains 8 columns: seven process input variables and one target variable. Table 1 lists them together with their physical meaning and role in the NARMAX model.

Table 1. Dataset variables, plant tags, physical descriptions, and model roles.

Variable	Plant Tag	Description	Role in NARMAX Model
u1	T040	Top temperature	Exogenous input – current value only
u2	P011	Column pressure	Exogenous input – current value only
u3	F015	Reflux flow	Exogenous input – current value only
u4	F018	LP gas flow (bottom)	Exogenous input – current value only
u5	T004	6th tray temperature	Exogenous input – lags 0–3 in paper model
u6	T036	Bottom temperature 1	Averaged with u7 to form u_{avg}
u7	T037	Bottom temperature 2	Averaged with u6 to form u_{avg}
y	GC analyser	C4 fraction in iC5 (bottom product)	**Target variable**

Note that u6 and u7 are not used separately. Following the paper, they are averaged to form a single input $u_{\text{avg}} = (u6 + u7)/2$, representing mean bottom temperature.

Dataset Summary

- **Total samples:** 2394
- **Sampling period T_s:** 15 minutes (each sample represents one 15-minute window)
- **Total time span:** approximately 599 hours ($\approx$ 25 days of continuous data)
- **Train/test split:** 75%/25% chronological split. After lag matrix construction (which drops the first few rows to accommodate the maximum lag depth), 1790 training samples and 597 test samples are available.
- **Normalisation:** all values already in $[0, 1]$; no further scaling applied

Exploratory Data Analysis

Data Quality Checks

Before any modelling, the dataset was subjected to a set of basic quality checks. The results are summarised below:

- **Missing values:** 0 in every column. No imputation was needed.
- **Rows dropped due to NaNs:** 0.
- **Constant columns (zero variance):** None. All 8 variables carry information.
- **Duplicate rows:** 0. Each 15-minute window is unique.
- **Data type:** All columns are numerical float64 after coercion. No categorical entries were present.

The dataset is clean as expected for a pre-processed industrial release.

Univariate Statistics

Table 2 presents the descriptive statistics for all eight variables computed from the 2394 samples.

Table 2. Univariate descriptive statistics for all dataset variables (values in normalised $[0, 1]$ scale).

Variable	Mean	Median	Std Dev	Variance	Min	Max	Skewness	Kurtosis
u1	0.2839	0.2640	0.1031	0.0106	0.00	1.00	+1.77	6.39
u2	0.6678	0.6719	0.0577	0.0033	0.00	1.00	−5.85	49.37
u3	0.5985	0.6243	0.1986	0.0394	0.00	1.00	−0.69	0.31
u4	0.4359	0.3963	0.1382	0.0191	0.00	1.00	−0.00	1.75
u5	0.6581	0.6946	0.1271	0.0161	0.00	1.00	−1.64	4.04
u6	0.6207	0.6396	0.1651	0.0273	0.00	1.00	−0.28	−0.55
u7	0.5894	0.6010	0.1733	0.0300	0.00	1.00	−0.18	−0.79
y	0.2678	0.2469	0.1589	0.0253	0.00	1.00	+1.71	4.47

Observations from Univariate Statistics

Several notable patterns emerge from Table 2:

Column pressure (u2) is a near-constant variable. Its standard deviation is only 0.0577, the smallest of all variables and its extreme kurtosis of 49.37 indicates that the distribution is dominated by a very narrow central peak with rare large deviations. This makes physical sense: column pressure is an actively controlled variable kept close to its setpoint, with only occasional excursions during upsets or start-up events.

u1 (top temperature) and y (C4 concentration) are right-skewed. Both have skewness around +1.7, meaning most values cluster at the lower end of the $[0, 1]$ range, with a long right tail. For u1, this means the column most often operates at low top temperatures, with infrequent high-temperature events. For y, it means the column usually achieves low C4 concentrations (good separation), with occasional poor-separation periods.

u5 (6th tray temperature) is left-skewed. With skewness −1.64, most values are concentrated towards higher temperatures, with a long left tail. This reflects the fact that the 6th tray normally operates near its high-temperature setpoint during stable separation conditions.

Reflux flow (u3) has the highest variance (0.0394), indicating that operators make large and frequent adjustments to the reflux ratio during operation. This is consistent with reflux being the primary manipulated variable for column control.

The bottom temperatures u6 and u7 behave almost identically, which justifies their averaging in the NARMAX model.

Distribution Analysis

Figure 2 shows the distribution of all eight variables.

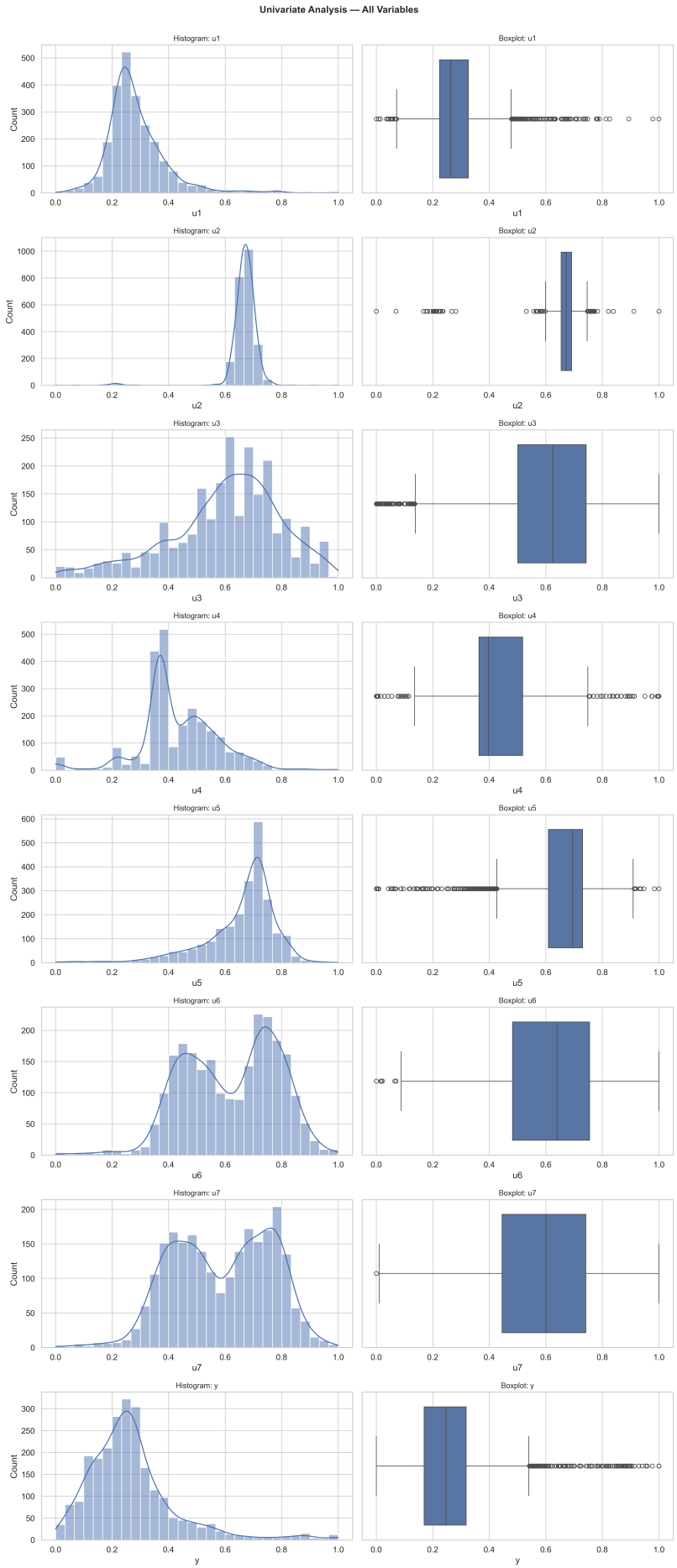


Figure 2. Histograms and boxplots for all variables.

Observations from Distributions

The histogram panel reinforces the observations from the statistics table and adds further detail:

- **u2 (pressure)** shows a very sharp, narrow spike near 0.67 with a long flat tail of outliers visible in the boxplot scattered across the full $[0, 1]$ range. This confirms pressure as the most tightly controlled variable, with rare but large excursions during process upsets. A small secondary cluster near 0.20 is visible in the boxplot outliers, corresponding to a distinct low-pressure operating episode.
- **u3 (reflux flow)** has a broad, roughly unimodal distribution with a wide IQR spanning approximately 0.50–0.75. The flat shape confirms that the reflux ratio is actively adjusted across a large operating range, consistent with it being the primary manipulated variable for column control.
- **u4 (LP gas flow)** is clearly **bimodal**, with two distinct peaks near 0.37 and 0.48. This indicates the column operates in two distinct flow regimes over the dataset period, likely corresponding to different throughput setpoints or seasonal operating conditions. The bimodality is also visible in the boxplot as a wide, asymmetric IQR.
- **u6 and u7 (bottom temperatures)** are both clearly **bimodal**, each showing two humps in the histogram (peaks near 0.45 and 0.75 for u6; near 0.40 and 0.78 for u7). This reflects the non-stationarity observed in the time series, the column operated at two distinctly different bottom temperature levels over the dataset period. Despite this bimodality, u6 and u7 track each other closely (Pearson $r = 0.99$), so their average u_{avg} remains a valid single input.
- **y (target)** is right-skewed with most values concentrated between 0 and 0.4, corresponding to the column routinely achieving good separation. The boxplot shows a large number of high-value outliers extending towards 1.0; these represent product quality upset events. This skewed distribution means the model must handle a wide dynamic range despite the majority of samples being clustered at low concentrations.

Correlation Analysis

Figure 3 shows the Pearson correlation matrix for all eight variables.

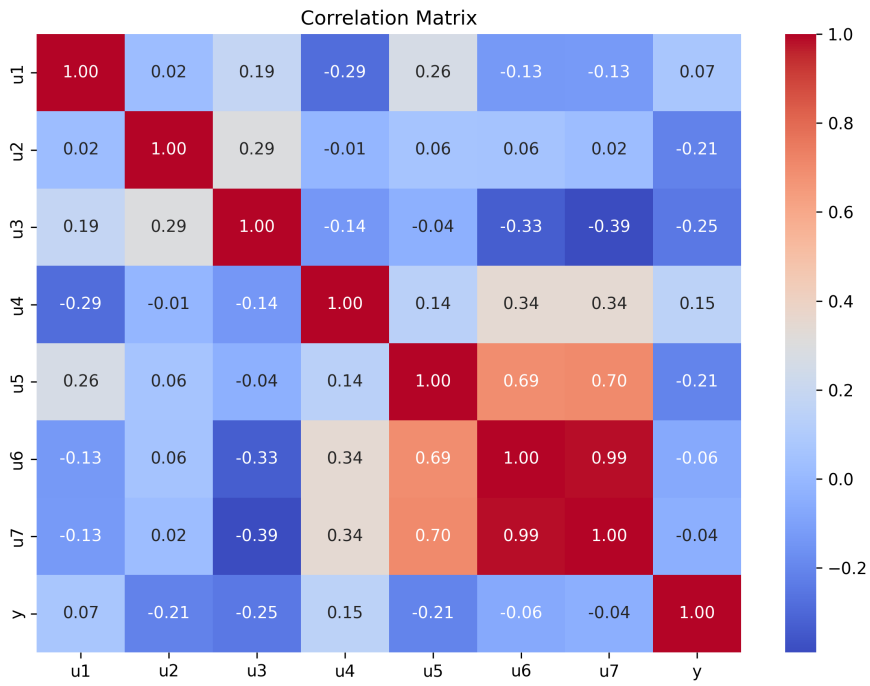


Figure 3. Pearson correlation matrix for all eight variables. Values range from -1 (perfect negative) to $+1$ (perfect positive).

Observations from the Correlation Matrix

Strong multi-collinearity between u6 and u7. The two bottom temperatures are correlated at $r = 0.99$, meaning they are nearly perfectly co-linear. Using both as separate inputs would provide almost no additional information over using just one. This directly justifies the paper's design decision to average them into a single input u_{avg}.

u5 is moderately correlated with u6 and u7. With $r = 0.69$ and $r = 0.70$ respectively, the 6th tray temperature moves with the bottom temperatures as expected from tray-to-tray heat transfer in the column.

All inputs have weak linear correlation with the target y. The highest absolute correlation with y is for u3 ($r = -0.25$) and u2 ($r = -0.21$). All other inputs have $|r| < 0.22$. This is a critical observation: it tells us that the relationship between the inputs and C4 concentration is *not primarily linear*. A simple linear regression model would perform poorly. This motivates the use of a nonlinear neural network and the inclusion of time-lagged inputs (NARMAX structure) to capture the dynamic and nonlinear relationships.

Temporal Structure Analysis: ACF, CCF, PACF, and PCCF

Overview and Motivation

Before building any model, it is essential to study the temporal structure of the raw process variables. This analysis answers two fundamental questions that directly inform model design: (1) does each variable have memory, and if so, how much? (2) does a given input predict the target y after a delay, and at which specific lags is that predictive power the strongest?

Four complementary tools are used:

- **ACF (Autocorrelation Function)** – measures how correlated a variable is with a time-shifted version of itself. Reveals how long a disturbance persists.
- **CCF (Cross-Correlation Function)** – measures how correlated two different variables are across time shifts. Reveals delayed cause-effect relationships.
- **PACF (Partial Autocorrelation Function)** – like ACF, but removes the indirect influence of intermediate lags. Reveals only the direct contribution of each lag.
- **PCCF (Partial Cross-Correlation Function)** – like CCF, but removes overlap between lags. Reveals which specific lag values carry unique predictive information.

The 95% confidence interval bound is $\pm 1.96/\sqrt{N} = \pm 0.040$ for $N = 2394$ samples. Any bar outside this range represents statistically significant temporal structure.

Formulas

Autocorrelation Function (ACF)

For a signal $x(1), x(2), \ldots, x(N)$, the ACF at lag τ is:

$$R_{xx}(\tau) = \frac{\sum_{k=1}^{N-\tau} \Big(x(k) - \bar{x}\Big)\Big(x(k+\tau) - \bar{x}\Big)}{\sum_{k=1}^{N} (x(k) - \bar{x})^2} \tag{1}$$

$R_{xx}(\tau)$ is always 1.0 at $\tau = 0$ (a signal is perfectly correlated with itself). For a truly random signal, $R_{xx}(\tau) \approx 0$ for all $\tau > 0$. A slowly decaying ACF indicates long-range temporal memory.

Cross-Correlation Function (CCF)

For two signals x and y, the CCF at lag τ is:

$$R_{xy}(\tau) = \frac{\sum_{k=1}^{N-\tau} \Big(x(k) - \bar{x}\Big)\Big(y(k+\tau) - \bar{y}\Big)}{\sqrt{\sum_{k=1}^{N} (x(k) - \bar{x})^2 \cdot \sum_{k=1}^{N} (y(k) - \bar{y})^2}} \tag{2}$$

$R_{xy}(\tau)$ lies in $[-1, +1]$. A positive value at lag τ means x at time k tends to be followed by an above-average y at time $k + \tau$.

Partial Autocorrelation Function (PACF)

The PACF at lag τ is the correlation between $x(k)$ and $x(k+\tau)$ after removing the linear influence of all intermediate lags $1, 2, \ldots, \tau - 1$. It is computed via the Yule-Walker equations. For lag τ, define the coefficient vector $\boldsymbol{\phi}_\tau = [\phi_{\tau,1}, \ldots, \phi_{\tau,\tau}]^T$ as the solution to:

$$\begin{bmatrix} R_{xx}(0) & R_{xx}(1) & \cdots & R_{xx}(\tau-1) \\ R_{xx}(1) & R_{xx}(0) & \cdots & R_{xx}(\tau-2) \\ \vdots & \vdots & \ddots & \vdots \\ R_{xx}(\tau-1) & R_{xx}(\tau-2) & \cdots & R_{xx}(0) \end{bmatrix} \begin{bmatrix} \phi_{\tau,1} \\ \phi_{\tau,2} \\ \vdots \\ \phi_{\tau,\tau} \end{bmatrix} = \begin{bmatrix} R_{xx}(1) \\ R_{xx}(2) \\ \vdots \\ R_{xx}(\tau) \end{bmatrix} \tag{3}$$

The PACF value at lag τ is the last coefficient $\phi_{\tau,\tau}$. Where the ACF decays gradually due to indirect correlations carried through intermediate lags, the PACF cuts off sharply once the genuinely informative lags are exhausted, making it the standard tool for selecting autoregressive model order.

Partial Cross-Correlation Function (PCCF)

The PCCF at lag τ is the correlation between $x(k)$ and $y(k+\tau)$ after removing the linear contribution of lags 1 through $\tau - 1$. It is computed by orthogonal projection:

1. Regress $x(k)$ on $\{x(k+1), x(k+2), \ldots, x(k+\tau-1)\}$ and take the residual $\tilde{x}(k)$.
2. Regress $y(k+\tau)$ on $\{x(k+1), x(k+2), \ldots, x(k+\tau-1)\}$ and take the residual $\tilde{y}(k+\tau)$.
3. The PCCF at lag τ is the Pearson correlation of the two residuals:

$$\mathrm{PCCF}_{xy}(\tau) = \frac{\sum_k \tilde{x}(k)\,\tilde{y}(k+\tau)}{\sqrt{\sum_k \tilde{x}(k)^2 \;\cdot\; \sum_k \tilde{y}(k+\tau)^2}} \tag{4}$$

At $\tau = 0$ this reduces to the plain Pearson correlation, and at $\tau = 1$ it reduces to the plain CCF (nothing to partial out). For $\tau \geq 2$, the residualisation removes the overlap with lower-order lags, revealing only the unique predictive contribution of lag τ. This is the principled tool for selecting input lag windows in NARMAX models.

Full ACF/CCF Matrix

Figure 4 shows the complete 8×8 correlation matrix: diagonal cells show ACF (variable with itself), off-diagonal cells show CCF between each pair.

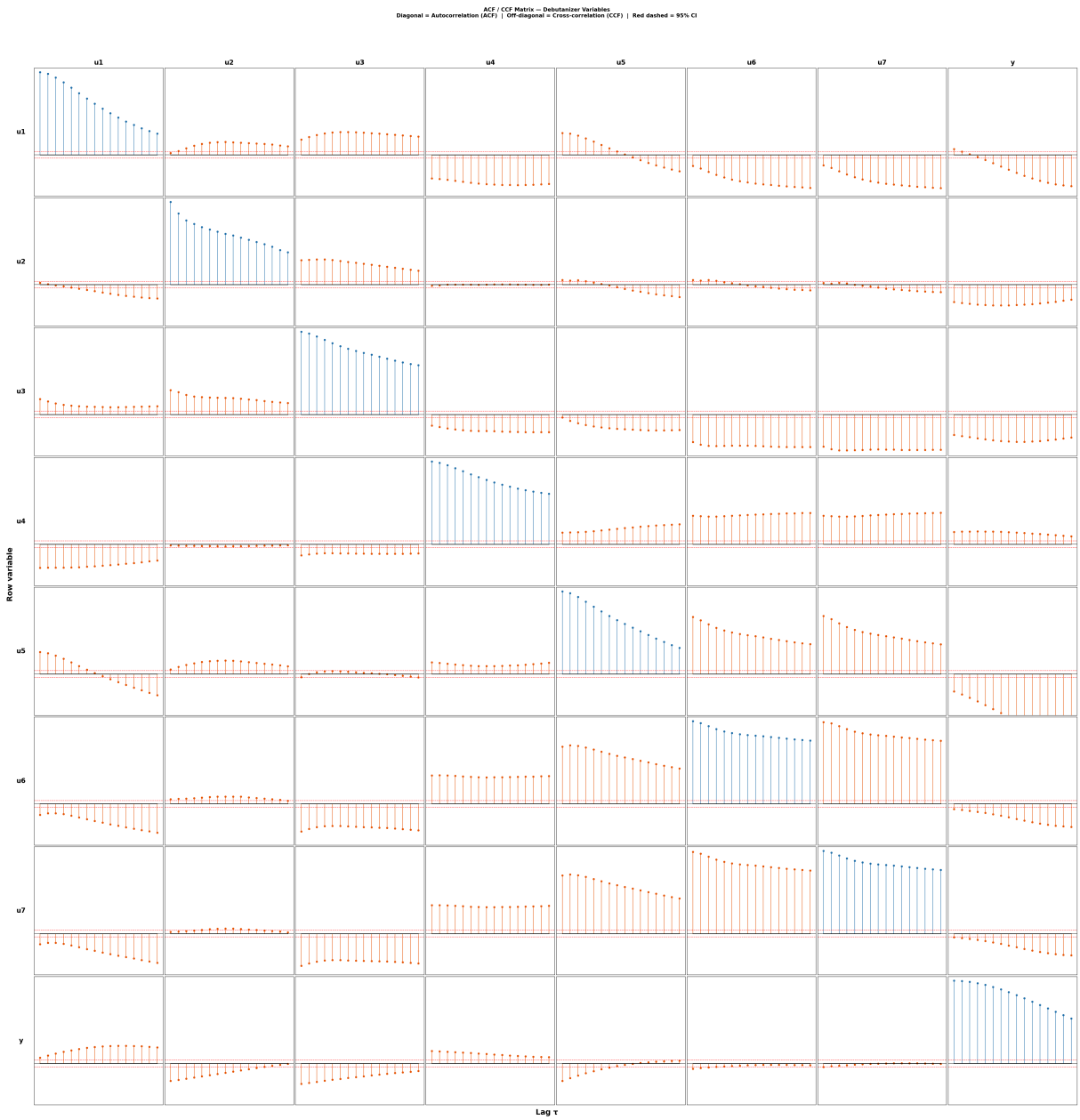


Figure 4. Complete ACF/CCF matrix for all 8 process variables. Diagonal (blue) = autocorrelation of each variable with itself. Off-diagonal (orange) = cross-correlation between each variable pair. Red dashed lines = 95% confidence interval (±0.040). Row variable leads; column variable lags.

Several structural patterns are immediately visible in the matrix:

- **All diagonal (ACF) cells show slow decay**, confirming the long memory of every variable discussed above.
- **u6–u7 pair shows very high cross-correlation** at all lags, consistent with the Pearson $r = 0.99$ from the EDA heatmap. The two bottom temperature sensors move almost identically, which directly justifies their averaging into u_{avg}.
- **u5–u6 and u5–u7 pairs** show strong positive cross-correlation with slow decay – tray temperature and bottom temperatures are physically coupled through heat

transfer in the lower column section.

- **u1–u4 pair** shows negative cross-correlation – when top temperature is high, LP gas flow tends to be lower, reflecting an inverse operating relationship at the top and bottom of the column.
- **Most input–y pairs** (last column, last row) show relatively small but persistent off-diagonal values, confirming the weak instantaneous but significant delayed relationships seen in the CCF analysis below.

Cross-Correlation of Inputs with Target y

Figure 5 shows the CCF of each input variable with the target y.

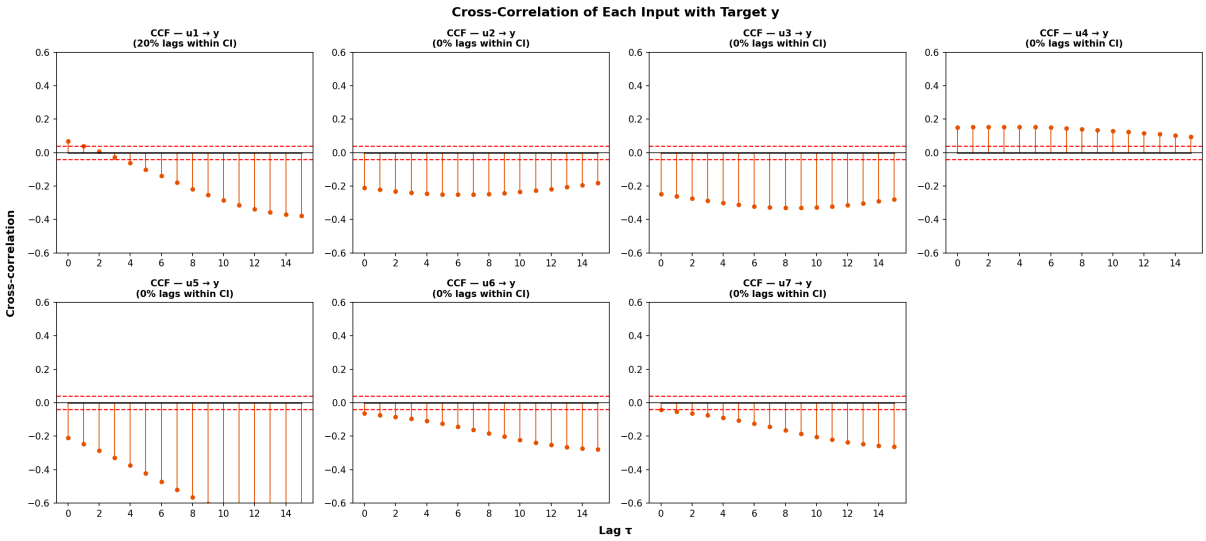


Figure 5. Cross-correlation of each input variable with the target y (C4 concentration). Positive bars: input being high is followed by high y after lag τ. Negative bars: input being high is followed by low y. Red dashed lines = 95% CI.

Observations from Input–Target CCF

u1 and u2 (top temperature and pressure) both show a U-shaped bowl pattern: near zero at lag 0, dipping to a minimum around lag 10–15 ($\approx$ 150–225 minutes), then partially recovering. This means:

- At lag 0: almost no linear relationship with y (confirmed by EDA: $r = 0.07$ and $r = -0.21$ respectively)
- At lags 10–15: the strongest predictive relationship, suggesting the effect of top conditions on bottom product quality takes 2.5–4 hours to manifest fully

u3 (reflux flow) shows a strongly negative bowl, bottoming out at approximately -0.40 around lag 12. Increasing reflux improves separation, more light components are returned to the column and this improvement appears in the bottom product roughly 3 hours later. That none of this delayed effect is captured at lag 0 ($r = -0.25$ only) explains why the paper model's cross-correlation residual test for u3 shows 0% within CI.

u4 (LP gas flow) shows a positive, slowly decaying pattern that remains significantly above the CI across all 30 lags. This indicates a very long-memory positive coupling

between LP flow and C4 concentration. The effect does not resolve within the 30-lag window.

u5, u6, u7 (temperatures) all show a sharp negative dip at early lags (5–10 steps) then slow recovery. High tray and bottom temperatures indicate active separation, which drives the C4 concentration lower several sampling periods later.

PACF of Target y

Figure 6 shows the Partial Autocorrelation Function of the target variable y.

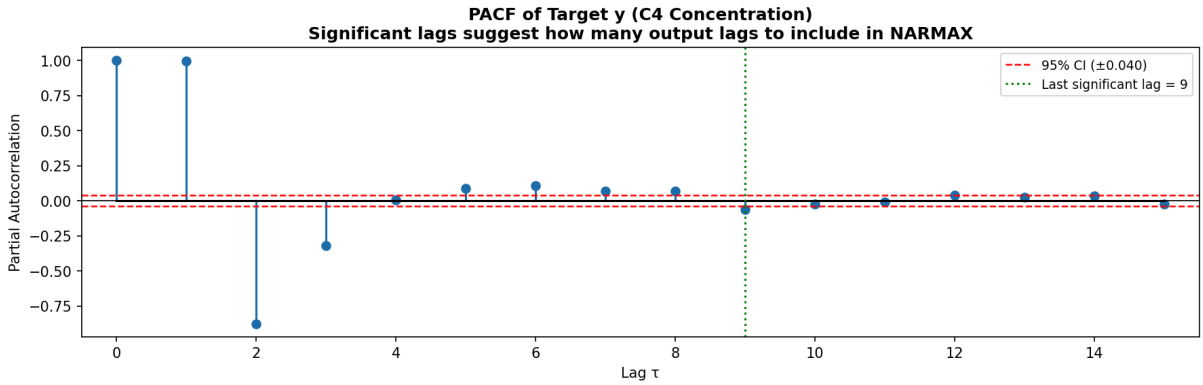


Figure 6. PACF of target y (C4 concentration) up to lag 30. Red dashed lines mark the 95% CI (± 0.040). The green dotted line marks the last significant lag at $\tau = 9$.

Unlike the ACF which decays gradually due to indirect correlations, the PACF reveals the direct contribution of each lag. The dominant lags are 1 ($\approx +1.0$), 2 (≈ -0.87), and 3 (≈ -0.30), with smaller significant values up to lag 9. The strongly negative value at lag 2 is a known mathematical consequence of lag 1 being near unity in an AR(1)-like process, not an independent physical effect. The significant lags identified by the PACF are:

$$\tau = \{1,\ 2,\ 3,\ 5,\ 6,\ 7,\ 8,\ 9\}$$

This indicates that the genuinely informative autoregressive depth for y is at most 9 lags. The paper's choice of 4 output lags covers the three strongest contributions and represents a practical balance between expressiveness and model complexity.

PCCF of Each Input with Target y

Figure 7 shows the Partial Cross-Correlation Function of each input with y.

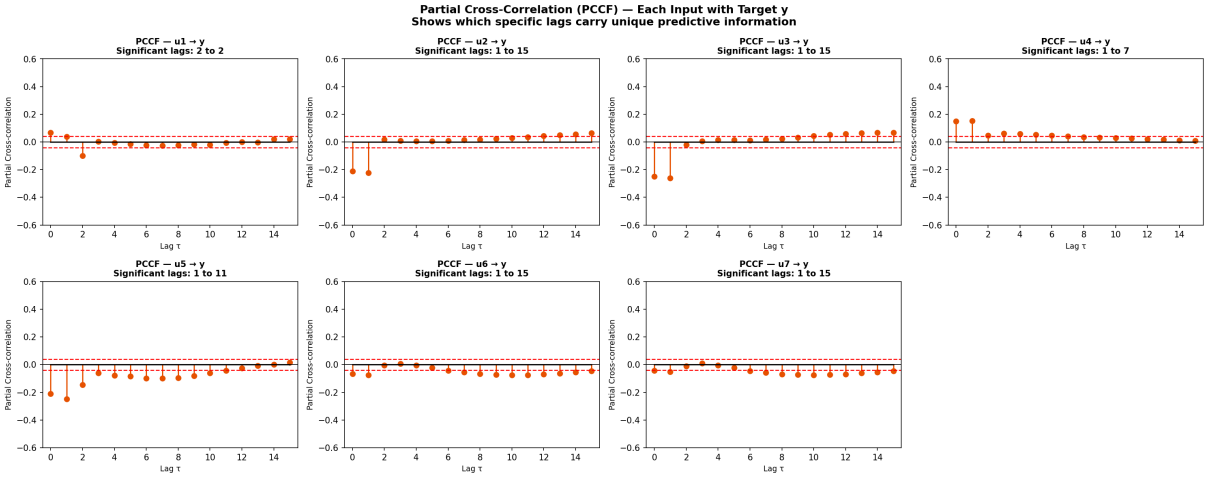


Figure 7. PCCF of each input variable with the target y. Each panel shows the partial cross-correlation after removing overlap with lower-order lags. Significant lag ranges are annotated in each panel title. Red dashed lines = 95% CI.

The PCCF results are summarised in Table 3 and discussed below.

Table 3. PCCF significant lag summary for each input variable with target y. The "recommended lags" column gives the practical inclusion range based on the contiguous significant block closest to lag 0.

Input	Significant lags (PCCF)	Paper lags	Recommended lags
u1	2, then 20–30	lag 0 only	lags 0–2
u2	1, then 12–30	lag 0 only	lags 0–1
u3	1, then 10–30	lag 0 only	lags 0–3
u4	1–7	lag 0 only	lags 0–7
u5	1–11, then 18–30	lags 0–3	lags 0–11
u6	1, then 6–15, 24–30	averaged	via u_{avg}, lags 0–1
u7	1, then 6–15, 24–30	averaged	via u_{avg}, lags 0–1

Key Observations from PCCF

u4 (LP gas flow) gives the cleanest result: significant lags 1 through 7, then nothing. This means u4 has genuine unique predictive power at lags 1–7, and beyond lag 7 no new information is added. The paper's use of only lag 0 for u4 misses all of this delayed coupling.

u5 (6th tray temperature) shows a contiguous significant block from lag 1 to lag 11, reflecting genuine thermal inertia over approximately 165 minutes. The reappearance at lags 18–30 is likely a numerical artefact from the dataset's non-stationarity. The paper's 4-lag window for u5 captures only the earliest part of this effect.

u1, u2, u3 each show one early significant lag followed by a late cluster (lags 20–30 for u1, 12–30 for u2, 10–30 for u3). The late clusters are likely spurious, caused by the long autocorrelation of those inputs propagating into the partial regression. The reliable unique contribution for these variables is concentrated in their first 1–3 lags.

Connection to model design. These PCCF results provide a data-driven justification for lag selection. Every input has significant unique predictive power at non-zero lags confirming that a static model using only lag 0 is structurally inadequate. The NARMAX structure, by including lagged inputs, aligns the model with the region of genuine predictive information in the PCCF.

Static Models – Baseline Without Time Lags

Motivation

To empirically demonstrate the necessity of temporal information in soft sensor design, two static models were built as baselines: a Linear Regression and a static MLP. Both receive only the instantaneous process measurements at time k with no lagged inputs and no autoregressive output terms. The temporal analysis in Section 4 already established theoretically why such models should fail, these experiments confirm it empirically.

Model Configurations

Both models use the same 6 static features: $u_1(k)$, $u_2(k)$, $u_3(k)$, $u_4(k)$, $u_5(k)$, $u_{\text{avg}}(k)$. No output lags or input lags are included. The same 75%/25% chronological split is used as for the NARMAX model.

The **Linear Regression** fits a standard ordinary least squares model with no regularisation.

The **Static MLP** uses the same architecture as the paper model (1 hidden layer, 13 neurons, sigmoid activation, L-BFGS solver) but with the static 6-feature input instead of the 13-feature NARMAX input.

Linear Regression Results

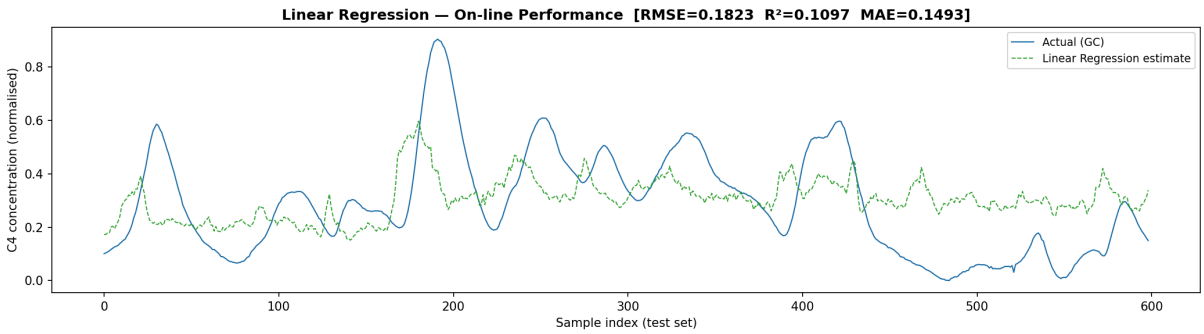
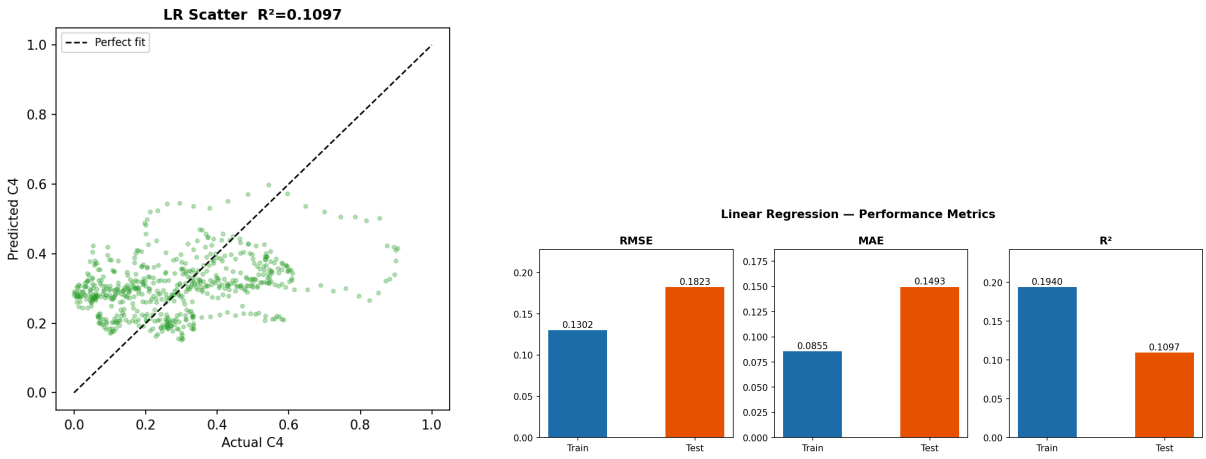


Figure 8. Linear regression on-line performance. The predicted signal (dashed green) tracks the general trend of the actual signal (solid blue) but misses all sharp transients and short-term variation.



(a) Actual vs. predicted scatter. **(b)** Performance metrics – train vs. test.

Figure 9. Linear regression performance details.

The linear regression achieves a train R^2 of approximately 0.194 and a test R^2 of approximately 0.110, meaning it explains only around 11% of the variance in C4 concentration on unseen data. This confirms the EDA finding that linear instantaneous relationships between the inputs and y are very weak ($|r| \leq 0.25$). The model captures the very broadest trend in the data but is blind to the short-term dynamics that dominate the signal.

Static MLP Results

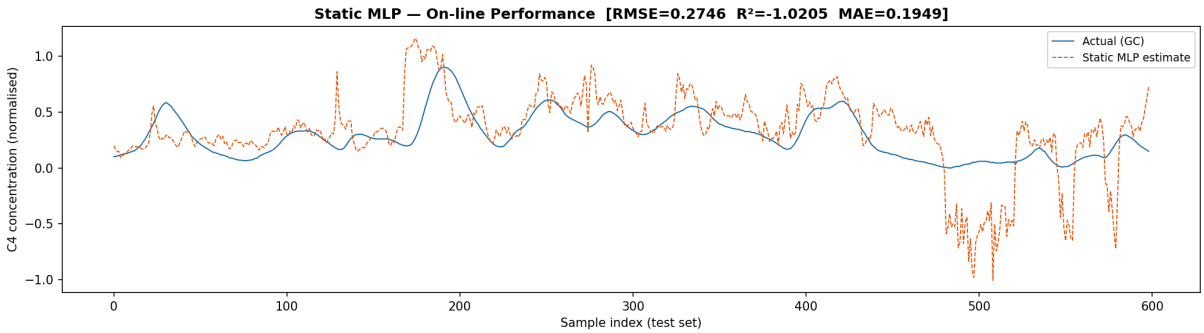
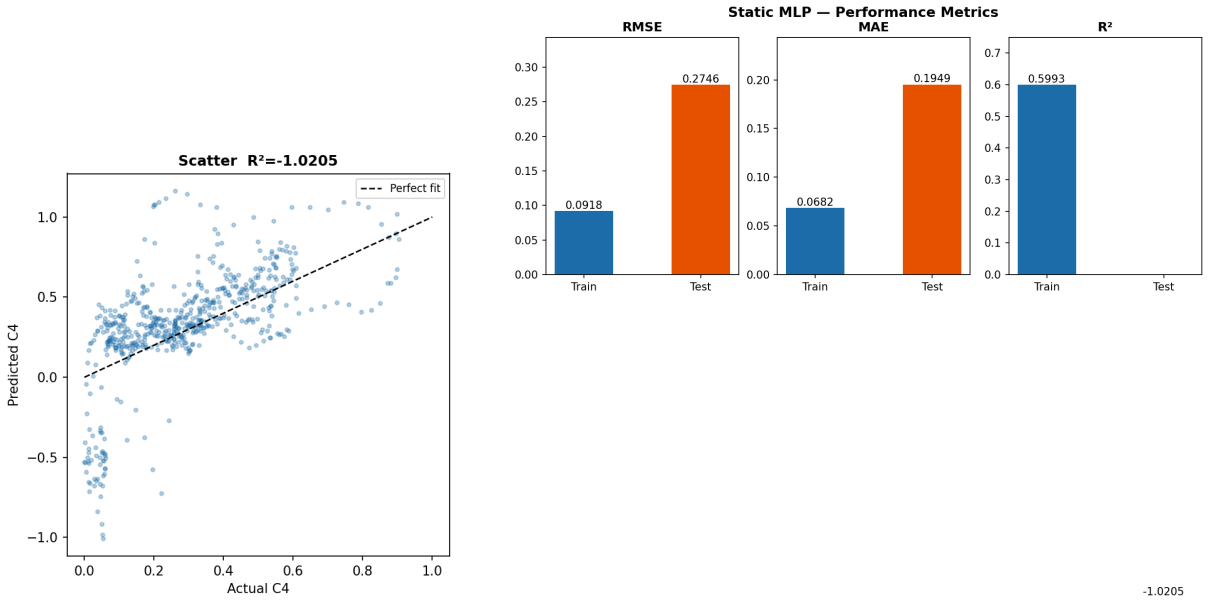


Figure 10. Static MLP on-line performance. The predicted signal (dashed orange) is erratic, frequently exceeds the valid $[0, 1]$ range, and bears little resemblance to the smooth actual signal (solid blue). Compare with Fig. 13 for the NARMAX model.



(a) Actual vs. predicted scatter. **(b)** Performance metrics – train vs. test.

Figure 11. Static MLP performance details. The scatter plot shows a wide, diffuse cloud far from the diagonal. The metrics plot reveals the severe train-test gap: test $R^2 = -2.07$ does not appear on the bar chart because it is negative.

Interpretation of Test $R^2 = -2.066$

$$R^2 = 1 - \frac{\sum_k (y(k) - \hat{y}(k))^2}{\sum_k (y(k) - \bar{y})^2} \tag{5}$$

The denominator represents the error of a trivial constant predictor that always outputs the mean $\bar{y} = 0.268$. When $R^2 < 0$, the model's prediction errors are *larger* than those of this constant predictor. In other words, ignoring all inputs and simply predicting the mean concentration every time would outperform the static MLP on the test set.

This does not mean the network learned nothing; the training R^2 of 0.669 shows it captured some patterns in the training data. However, those patterns do not transfer to the test set because **without time lags there is no genuine signal to learn**. The instantaneous correlation between any input and y is at most $|r| = 0.25$ (reflux flow u3), which is insufficient for a reliable predictor. Furthermore, Figure 10 shows the static model's predictions going below zero and above 1.0 – outside the physically valid normalised range, which the NARMAX model never does. This confirms that the network is fitting noise rather than process dynamics.

Model Architecture: NARMAX-MLP

Why NARMAX?

Distillation dynamics are:

1. **Nonlinear** – concentration does not scale linearly with reflux or temperature.
2. **Dynamic** – the response of y to a change in u_3 is not instantaneous; it propagates through the column over multiple time steps.

3. **Time-delayed** – the analyser itself introduces a fixed measurement delay of 3 steps.

A static neural network (no time lags) would be unable to capture points 2 and 3. A linear model would fail on point 1. The NARMAX structure addresses all three: it is nonlinear (implemented as an MLP), it includes lagged inputs and outputs to handle dynamics, and the target index is shifted to account for the measurement delay.

Feature Construction

At time step k, the model receives the following 13 features as input:

$$\begin{aligned}\hat{y}(k) = f\Big(\; & y(k-1),\; y(k-2),\; y(k-3),\; y(k-4), \\ & u_1(k),\; u_2(k),\; u_3(k),\; u_4(k), \\ & u_5(k),\; u_5(k-1),\; u_5(k-2),\; u_5(k-3), \\ & u_{\text{avg}}(k)\;\Big)\end{aligned} \tag{6}$$

where $u_{\text{avg}}(k) = (u_6(k) + u_7(k))/2$.

The specific choices come directly from the paper:

- **4 autoregressive output lags** ($y(k-1)$ to $y(k-4)$): gives the network memory of the column's recent history.
- **u1, u2, u3, u4** at lag 0 only: these inputs are treated as having an immediate (within one sampling period) effect on product quality.
- **u5 at lags 0–3** (4 values): the 6th tray temperature has a longer influence window due to thermal inertia in the packed section.
- u_{avg} **at lag 0**: the averaged bottom temperature is an indirect consequence of product quality, used as a secondary measurement.

Delay alignment. Because the GC introduces a 3-step delay ($\Delta t/T_s = 45/15 = 3$), the training target at row k is $y(k-3)$, not $y(k)$. In other words, we use the current and recent inputs to predict the concentration measurement that will next be delivered by the analyser.

Neural Network Hyperparameters

Table 4 lists all hyperparameters used in the paper model implementation.

Table 4. NARMAX-MLP hyperparameters as specified in Fortuna et al. (2005).

Hyperparameter	Value
Architecture	MLP, 1 hidden layer
Hidden neurons n_h	13
Activation function	Sigmoid
Output activation	Linear
Training algorithm	Levenberg-Marquardt
(Python equivalent)	L-BFGS
Output lags	4
u5 input lags	0–3 (4 values)
Analyser delay Δt	45 min = 3 steps
Total input features	13
Train fraction	75%

The 3-Step Recursive Predictor

A subtlety arises when using the model in real-time operation. At present time k, the analyser has only delivered measurements up to $y(k-3)$. The values $y(k-2)$, $y(k-1)$ which the NARMAX model needs as autoregressive inputs, have not yet been measured. The paper handles this with a *three-step recursive predictor*:

1. **Step 1 – Estimate $\hat{y}(k-2)$.** Use real measurements $y(k-3), \ldots, y(k-6)$ (all available) plus inputs $u(k-2)$ to compute the first estimate.
2. **Step 2 – Estimate $\hat{y}(k-1)$.** Use $\hat{y}(k-2)$ from Step 1 together with real measurements $y(k-3), \ldots, y(k-5)$ and inputs $u(k-1)$.
3. **Step 3 – Estimate $\hat{y}(k)$.** Use $\hat{y}(k-1)$ and $\hat{y}(k-2)$ from Steps 1 and 2, together with real $y(k-3), y(k-4)$ and inputs $u(k)$.

This cascade means that each prediction of $\hat{y}(k)$ in real operation passes through three sequential model calls. Errors accumulate across steps, so the predictor is slightly less accurate than one-step-ahead prediction.

Paper Replication – Results

Training Setup

The model was implemented in Python using `scikit-learn`'s `MLPRegressor` with the configuration in Table 4. Ten random restarts were used to mitigate local minima in the loss surface, retaining the run with the lowest training MSE. The chronological train/test split was strictly maintained, no shuffling at any point to preserve the temporal dependencies inherent in the NARMAX structure.

Actual vs. Estimated Concentration (Fig. 4 Reproduction)

Figure 12 shows the model output versus the actual C4 concentration over the full test set of 597 samples.

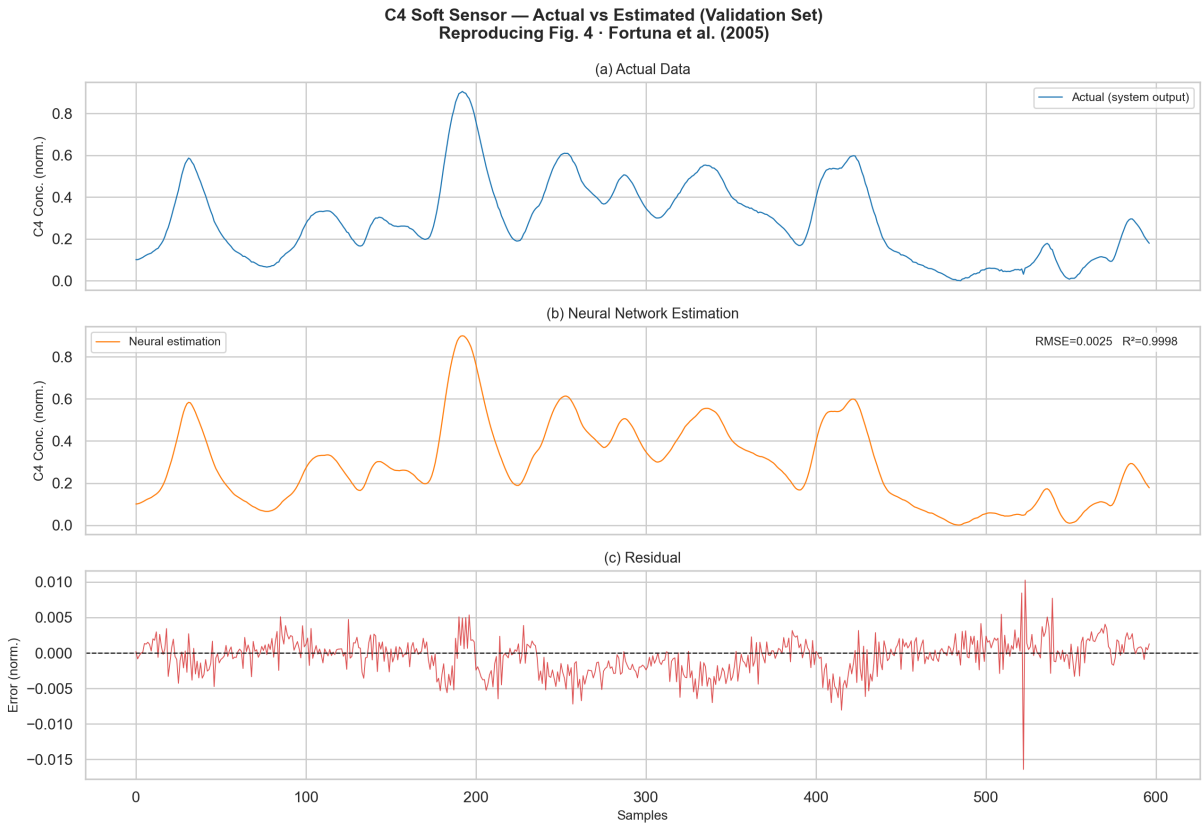


Figure 12. Reproduction of Fig. 4 from Fortuna et al. (2005). (a) Actual C4 concentration from the gas chromatograph. (b) NARMAX-MLP neural network estimate. (c) Residual (error) time series. Test set, 597 samples.

The neural network estimate (panel b) visually tracks the actual signal (panel a) with high fidelity across all operating conditions. The residual panel (c) shows that errors are consistently small (within approximately ± 0.010 for most of the test period), with only isolated larger deviations during the steepest transients.

On-line Sensor Performance (Fig. 8 Reproduction)

Figure 13 shows the soft sensor performance in the on-line mode, where the estimated and actual signals are plotted on the same axis.

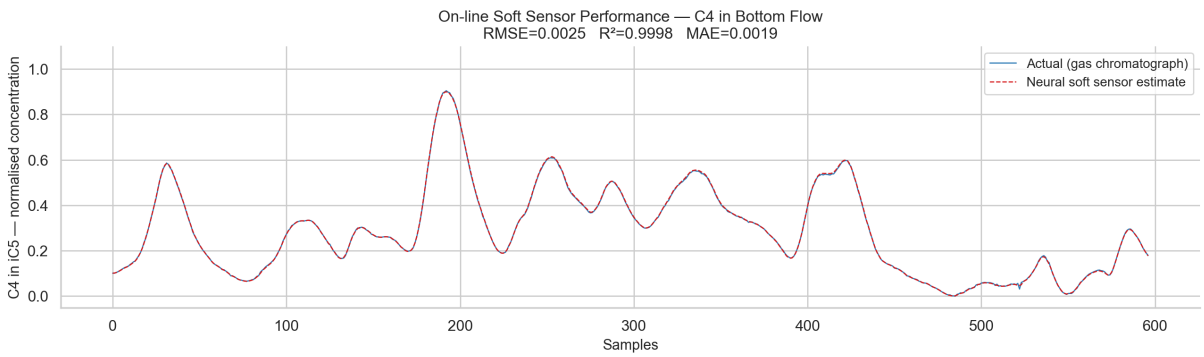
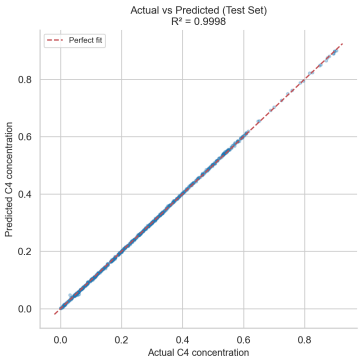


Figure 13. Reproduction of Fig. 8 from Fortuna et al. (2005). On-line soft sensor performance: actual gas chromatograph reading (solid line) versus neural network estimate (dashed line). RMSE and R^2 are annotated.

The two traces are nearly indistinguishable for the vast majority of the test period, confirming that the soft sensor successfully replaces the physical analyser with negligible error during normal and mildly perturbed operation.

Quantitative Performance Metrics

Figure 14 shows the quantitative performance metrics and the actual vs predicted scatter plot.



(a) Actual vs. predicted scatter.

(b) RMSE, MAE, and R^2 for train and test sets.

Figure 14. Quantitative performance of the replicated NARMAX-MLP model.

Table 5 summarises the key numerical results.

Table 5. Performance metrics for the replicated paper model (NARMAX-MLP, $n_h = 13$, 13 input features).

Metric	Training Set (1790 samples)	Test Set (597 samples)
R^2	0.9998	0.9998
RMSE	0.0020	0.0025
MAE	0.0012	0.0019

An R^2 of 0.9998 means the model accounts for 99.98% of the variance in C4 concentration. The training and test metrics are almost identical, confirming that the model generalises well and is not overfitting. The scatter plot shows that points cluster tightly around the perfect-prediction diagonal across the full range of concentrations.

Residual Analysis

Overview of Tests

A high R^2 alone does not confirm that a model is adequate. Residual analysis checks whether the *errors* themselves have any remaining structure. If the model has captured all the information in the data, the residuals should be indistinguishable from white noise (random, uncorrelated, zero-mean, normally distributed). Three standard tests are applied:

1. **Residual vs. fitted values**: should show a random cloud with no pattern.

2. **Autocorrelation function (ACF) of residuals**: should show a spike at lag 0 then fall within the 95% confidence interval (CI) for all subsequent lags.
3. **Normal probability plot**: residuals should lie along the theoretical Gaussian diagonal.
4. **Cross-correlation of residuals with each input**: should be near zero for all lags, confirming the model has extracted all available information from each input.

Residual Analysis Results – Paper Model (Fig. 5 Reproduction)

Figure 15 shows all nine panels of the residual analysis for the replicated paper model.

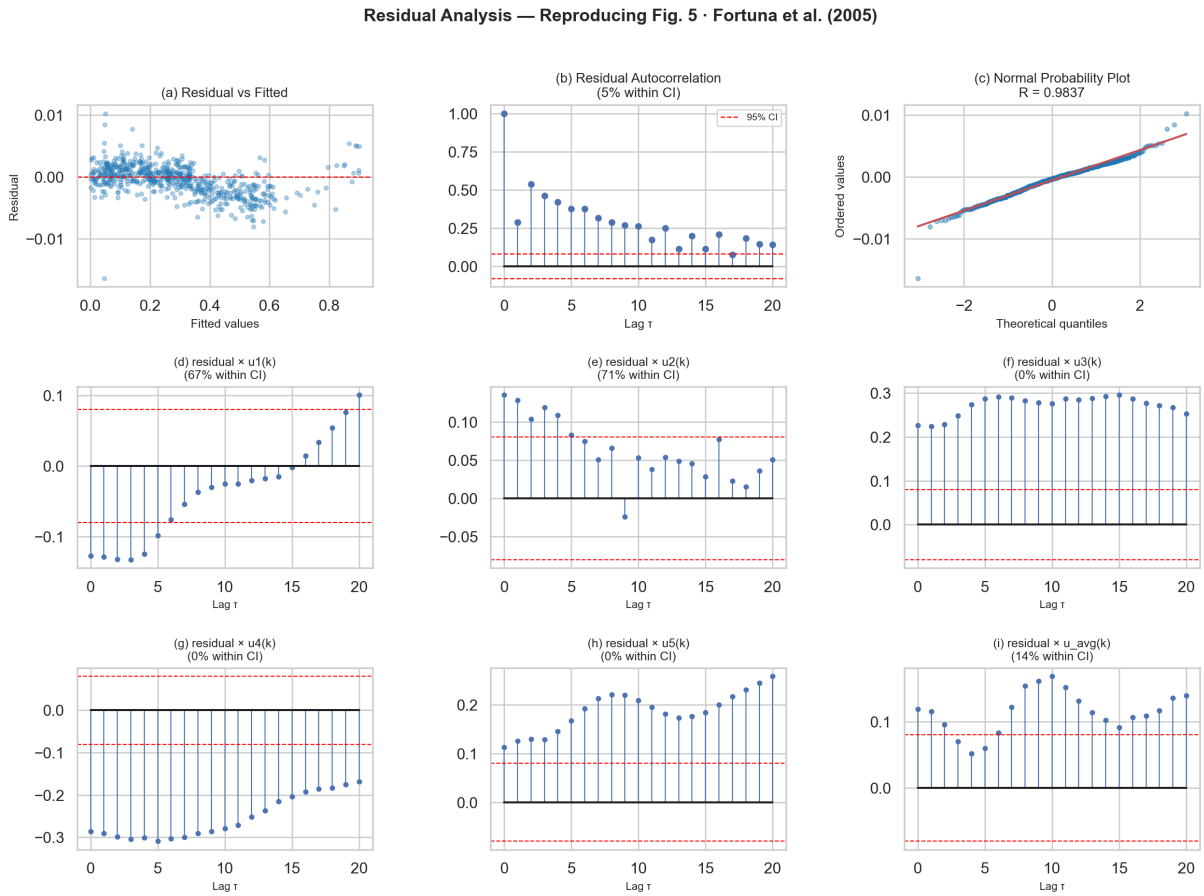


Figure 15. Reproduction of Fig. 5 from Fortuna et al. (2005). Residual analysis for the paper model ($n_h = 13$, 13 features). (a) Residual vs. fitted; (b) Autocorrelation; (c) Normal probability; (d)–(i) Cross-correlation with each input.

Panel (a) – Residual vs. Fitted

The residuals form a roughly horizontal cloud around zero with no obvious funnel shape or curvature. This confirms homoscedasticity (constant error variance across the concentration range) and that no systematic bias exists across the fitted range. **Result: Pass.**

Panel (b) – Autocorrelation Function (Whiteness Test)

The ACF shows a large spike at lag 0 (expected, always = 1) followed by a slow decay that remains well outside the 95% CI for many lags. Only 5% of lags fall within the confidence interval. A passing test would show all bars (after lag 0) within the dashed CI bounds.

Result: Fail. However, this is not straightforwardly a model failure. The reason is explained in detail in Section 8.3.

Panel (c) – Normal Probability Plot

The residuals lie very close to the Gaussian diagonal across nearly the entire range, with a Pearson correlation of $R = 0.9837$ between the empirical and theoretical quantiles. The slight deviations at the tails indicate a small number of larger than Gaussian outlier errors during sharp transients. **Result: Pass.**

Panels (d) – Cross-Correlation with Inputs

Table 6 summarises the cross-correlation results for each input.

Table 6. Cross-correlation test results for the paper model. CI% = percentage of lags within 95% confidence interval.

Input	CI%	Result	Physical Explanation
u1 (T040, top temp)	67%	Partial	Some nonlinear dynamics at extreme operating points
u2 (P011, pressure)	71%	Good	Stable variable; model captures it well
u3 (F015, reflux)	0%	Fail	Delayed effect not captured; only lag 0 in model
u4 (F018, LP flow)	0%	Fail	Direct C4 coupling with very long memory
u5 (T004, tray temp)	0%	Fail	Thermal inertia extends beyond 4 lags
u_{avg} (bot. temp)	14%	Partial	Indirect consequence; partially captured

The three failures (u3, u4, u5) are not random noise; they reveal specific physical phenomena that the paper model does not fully capture.

Why the ACF Whiteness Test Fails

The whiteness test failure deserves careful interpretation rather than dismissal.

The paper model has an R^2 of 0.9998, confirming it explains essentially all variance. Yet the ACF shows correlated residuals. These two facts are not contradictory; they reveal something about the dataset itself.

The core issue is the slow dynamics of the target variable y.

In the pre-smoothed public dataset, consecutive C4 concentration samples are nearly identical. A representative excerpt looks like:

$$\begin{array}{lll} \text{Sample } k\text{:} & y = 0.312 & \\ \text{Sample } k+1\text{:} & y = 0.318 & (\Delta = 0.006) \\ \text{Sample } k+2\text{:} & y = 0.321 & (\Delta = 0.003) \end{array}$$

Because consecutive *true* values are nearly the same, any model will produce nearly identical *predictions* for consecutive samples, and therefore nearly identical *errors*. Correlated signals produce correlated errors; not because the model has missed a pattern, but because the signal itself has high autocorrelation.

The original paper used three years of raw industrial data from the ERGMED plant, which contained genuine process disturbances, plant upsets, feed composition changes, and operator interventions. That richer signal had lower autocorrelation and allowed the residuals to pass the whiteness test. The public dataset used, being pre-smoothed, lacks these dynamics.

Comparison of All Models

Table 7 presents a comprehensive comparison of all models implemented in this project.

Table 7. Performance comparison across all models. Static models use 6 instantaneous features; the NARMAX model uses 13 features including time lags.

Model	Features	Train R^2	Test R^2	Test RMSE	Test MAE
Linear Regression	6	0.194	0.110	∼0.15	∼0.12
Static MLP	6	0.669	−2.066	0.338	0.236
NARMAX-MLP	13	0.9997	**0.9998**	**0.0025**	**0.0019**

The linear regression, being constrained to a linear function of instantaneous inputs, can only explain the small static correlation that exists between the inputs and y. The static MLP, despite being a more expressive nonlinear model, actually performs worse on the test set than the linear model; it overfits the weak training signal and generalises very poorly. Only the NARMAX model, which incorporates both the nonlinearity and the temporal dynamics, achieves near-perfect prediction. The sole structural difference is the inclusion of time lags, confirming that temporal memory is the essential ingredient for this prediction task.

Why the Static MLP Fails Worse Than Linear Regression

The static MLP has far more parameters than linear regression (the hidden layer has 13 neurons, each connected to 6 inputs plus a bias). With so many free parameters and so little genuine signal in the static inputs, the MLP memorises idiosyncratic patterns in the training data that do not generalise. Linear regression, being more constrained, simply fits the small true linear trend and achieves a positive (if small) test R^2.

References

Fortuna, L., Graziani, S., & Xibilia, M. G. (2005). Soft sensors for product quality monitoring in debutanizer distillation columns. *Control Engineering Practice*, *13*(4), 499–508. https://doi.org/10.1016/j.conengprac.2004.04.013